Politecnico di Torino

FDA: Fourier Domain Adaptation for Semantic Segmentation

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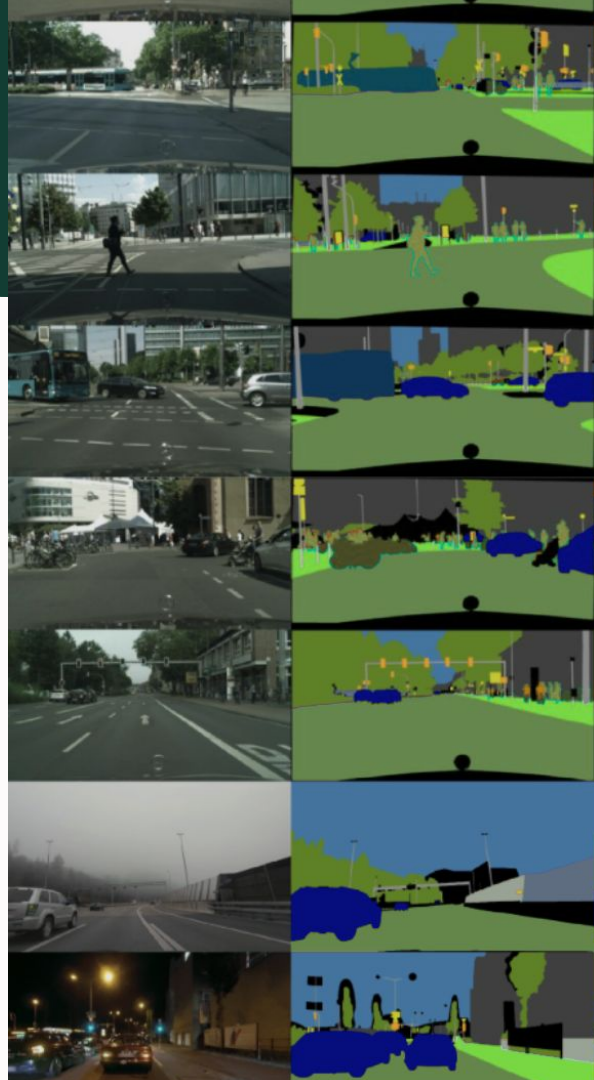
Overview

- Introduction
- State of the art
- The key insight
- How it works
- Advanced Training Strategies
- Datasets
- Results
- Discussion
- Conclusion



Introduction

Semantic segmentation works in one visual world, but often struggles when the style changes.

Training labels are expensive, so we often train on synthetic data and test on real images.

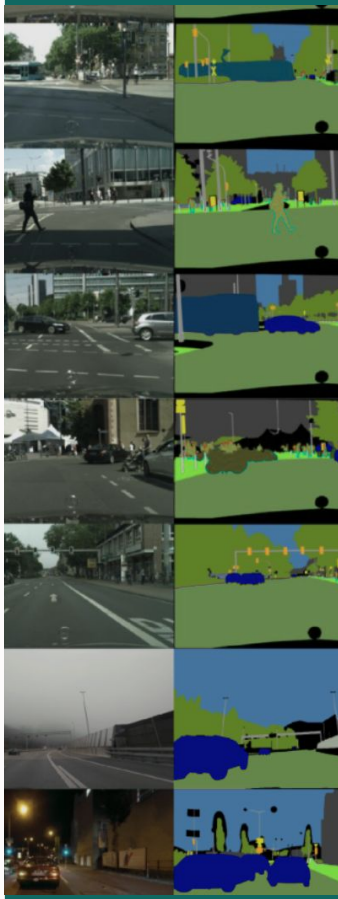
Semantic segmentation

- Assign a label to every pixel.
- Examples: road, car, pedestrian, building.
- Powerful for autonomous driving and scene understanding.

Why synthetic data?

- Pixel-level annotations are expensive to collect by hand.
- Synthetic datasets give labels automatically.
- The challenge appears when the target images look different.

The model learns the task in one visual world, and then struggles when that world changes.



The Domain Gap

The domain gap is mostly about style, not meaning.

The semantic content stays the same, but lighting, texture, color, and contrast can shift enough to break the model.

What stays the same

Road is still a road.
Car is still a car.
Building is still a building.

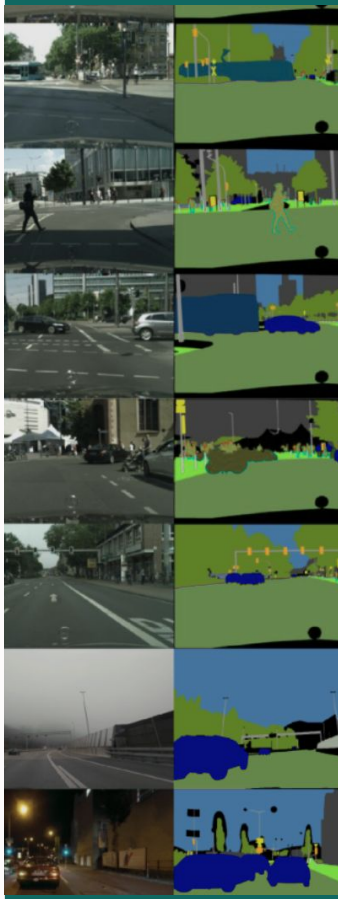
- The high-level structure stays stable.
- The model should focus on semantic content.

What changes

- **Lighting**
- **Contrast**
- **Texture and color statistics**
- **Camera / sensor style**

These are low-level appearance cues, not changes in meaning.

The mismatch is mostly about style, so the model should be as invariant as possible to style changes.



State Of The Art

SOTA

Most prior methods try to align the two domains inside the network.

The common idea is to close the domain gap by learning alignment, rather than removing the shift before training.

Adversarial alignment

- A discriminator tries to distinguish source and target domains.
- Useful, but the training game can be unstable.
- Dense prediction makes the balance even harder.

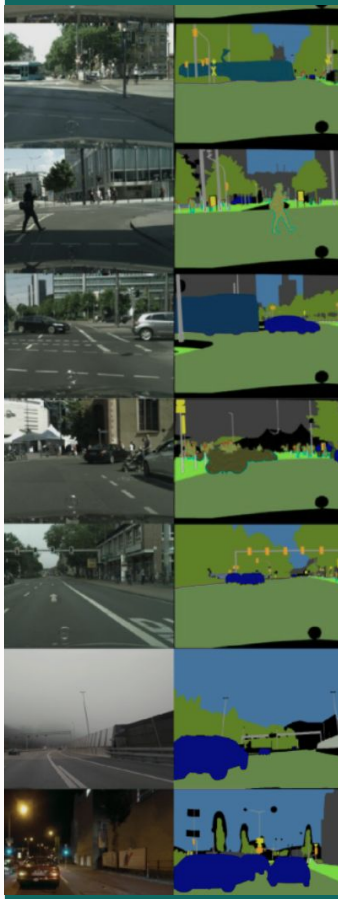
Image translation

- Translate source images to look like the target domain.
- Adds extra networks and another training stage.
- May alter details that segmentation still needs.

Feature/output alignment

- Match representations or prediction maps across domains.
- Works indirectly, so it depends on careful design.
- Still learns the style shift rather than removing it upfront.

Most solutions ask the network to learn away the shift.
FDA starts from a simpler idea: reduce the gap first.



Fourier Domain Adaptation

Source domain (synthetic)

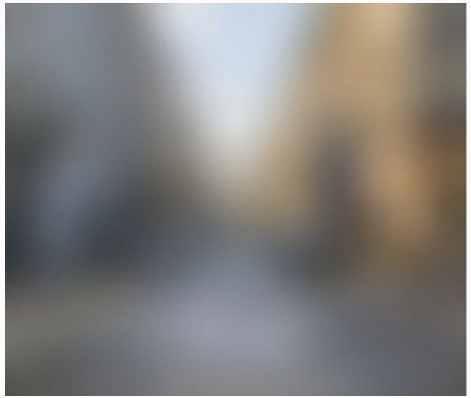
Covariate shift

Target domain (real)

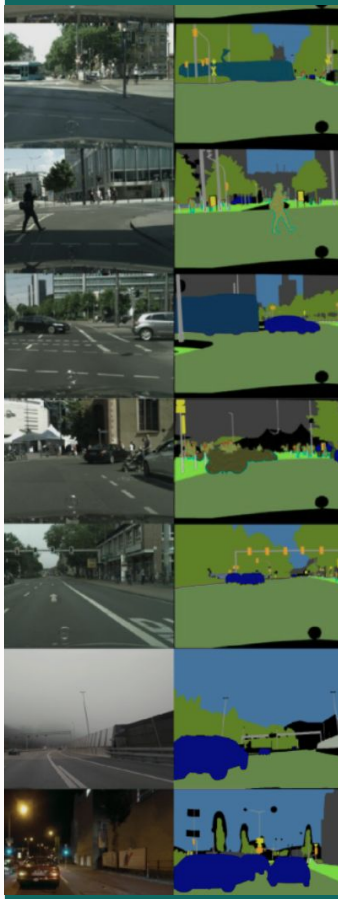
Phase: high-level Semantics (shapes-structure...)

The core theory

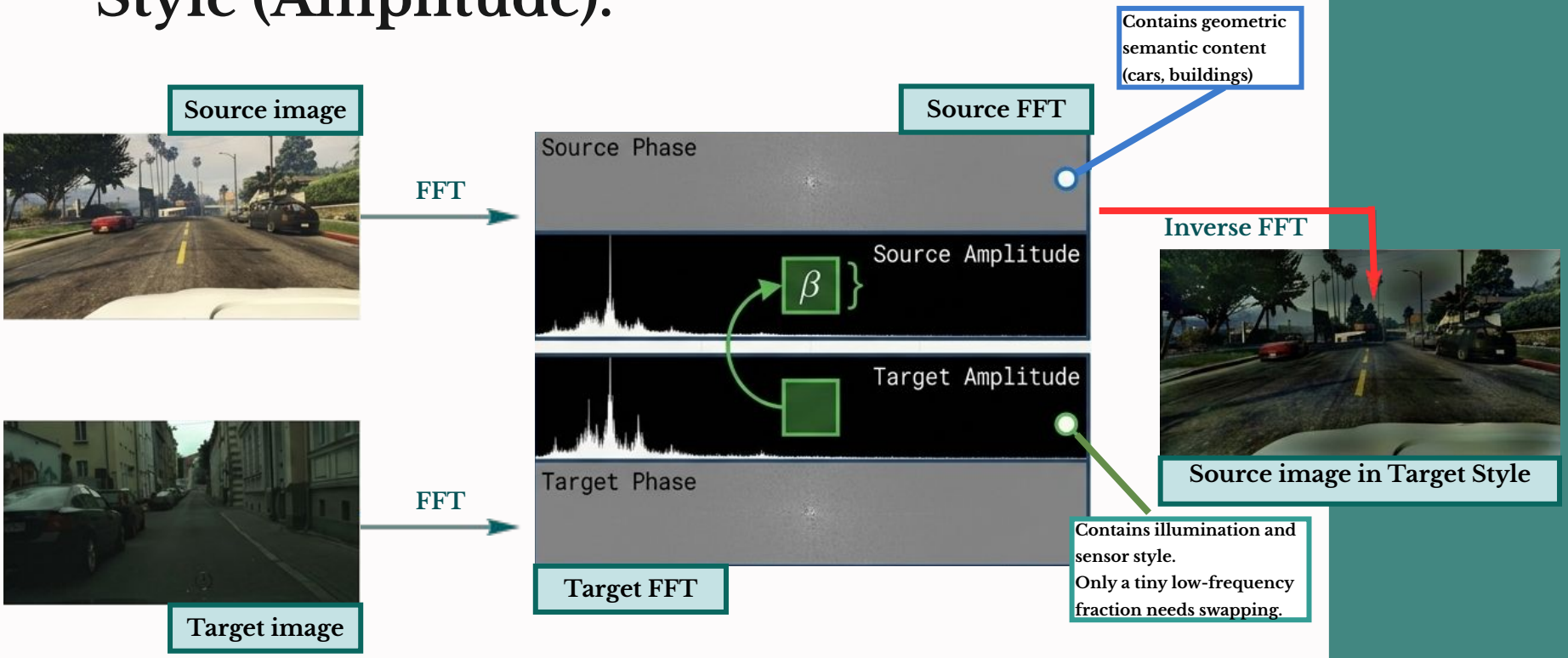
To fix the domain gap, we use the FDA that relies on the Fourier amplitude that changes a style without altering its semantic meaning.



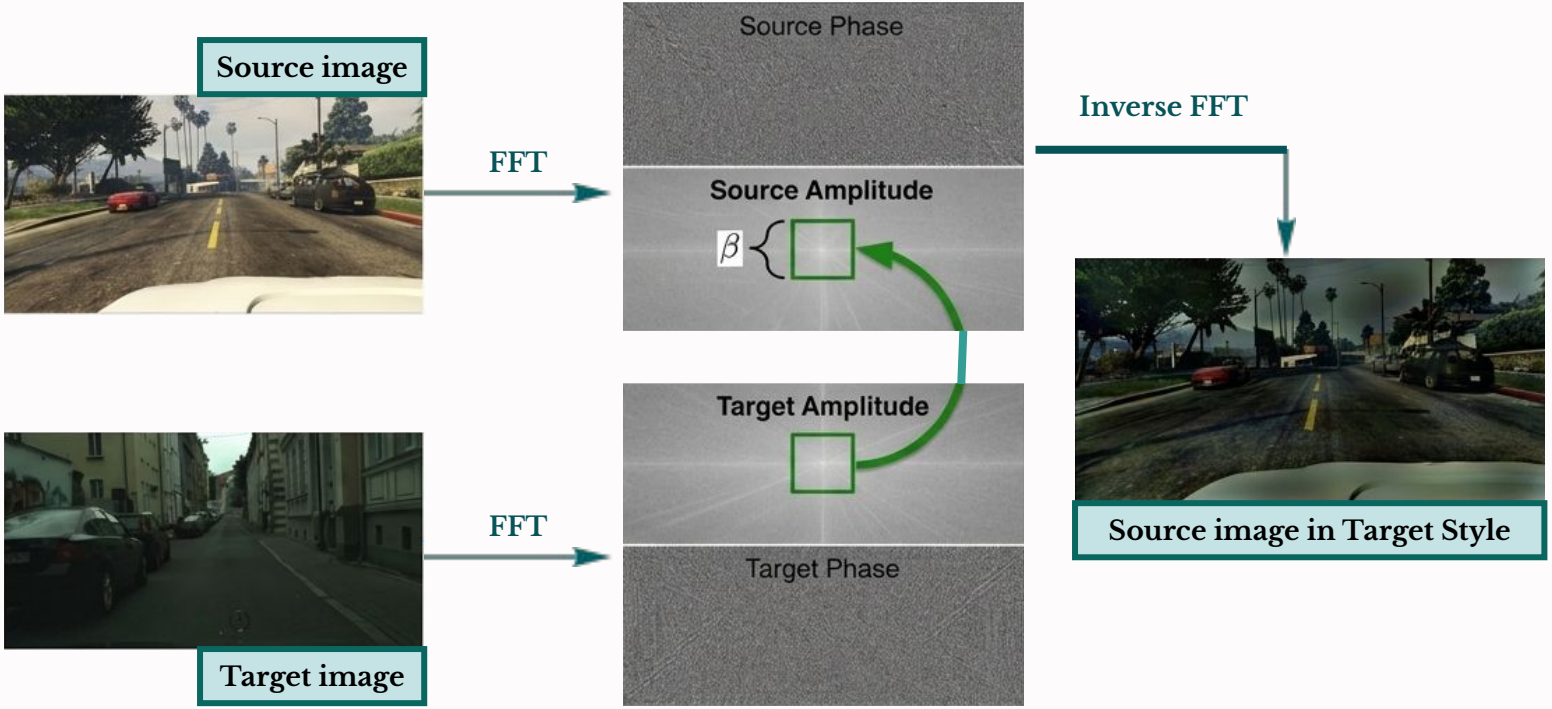
Amplitude: low-level Style (illumination, color...)

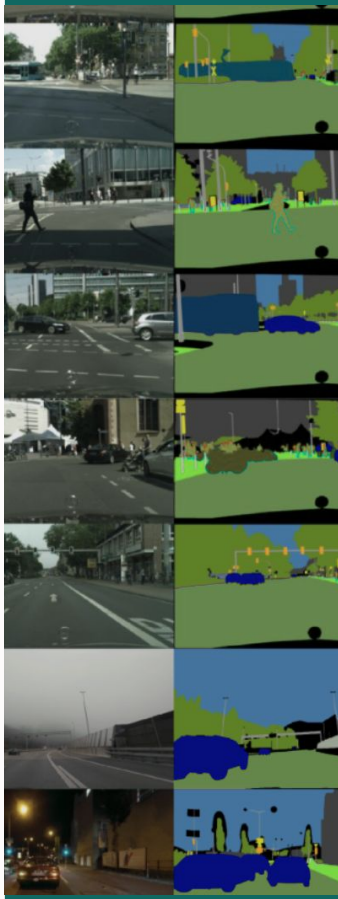


Separating Semantics (Phase) from Style (Amplitude):



The FDA Mechanism





The outcome preserves source semantics entirely while perceptually shrinking the domain gap to the target style.

01

Apply FFT to both source and target images

02

Swap low-frequency amplitude from the target into the source

03

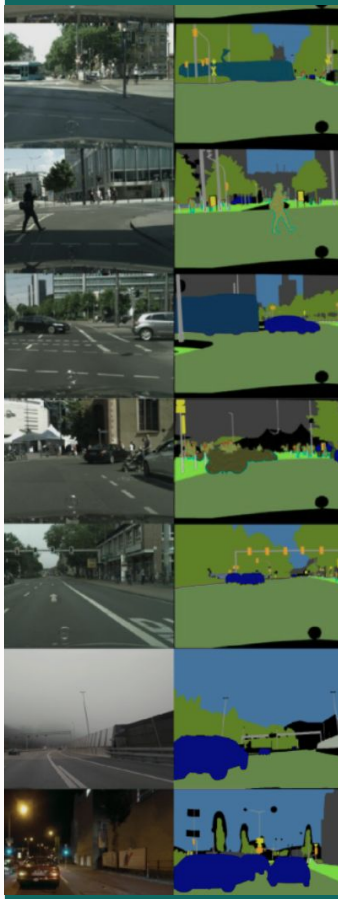
Apply Inverse FFT to generate the hybrid image

Insight

Because the phase of the source image is untouched, the ground-truth pixel labels from the synthetic data still perfectly match the object in the new, adapted image

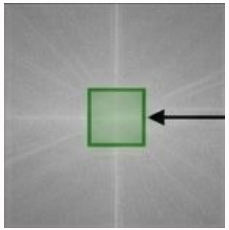
$$x^{s\to t} = \mathcal{F}^{-1}([M_\beta \circ F^A(x^t) + (1 - M_\beta) \circ F^A(x^s), F^P(x^s)])$$

- $x^{s\to t}$: The new adapted hybrid image.
- M_β: A mask that is zero everywhere except the center (low frequencies), controlled by $\beta \in (0, 1)$.
- F^A & F^P: The Amplitude and Phase components.
- $\mathcal{F}^{-1}$: The Inverse Fourier Transform.

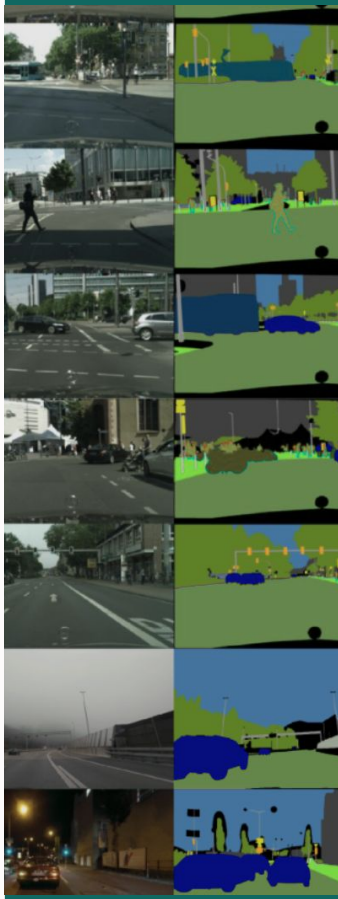


β=0.001 β=0.01 β=0.05 β=0.25

Source image



Mask (Mβ)



β ∈ (0,1) controls the size of the swapped low-frequency window in the amplitude spectrum.

The Trade-off:

We want β big enough to transfer the target domain's style, small enough to avoid severe visual artifacts (β <0.15)

FDA:

successfully aligns domains, using a simple, training-free mathematical swap

Advanced Training Strategies

Robust Entropy Minimization

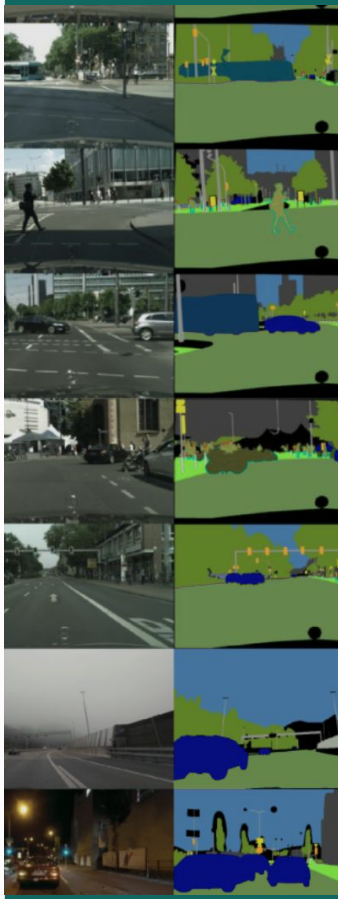
Unsupervised Domain Adaptation (UDA)

Fourier swapping

Semi-Supervised Learning (SSL)

Challenge

Regularization model



Robust Entropy Minimization

Unsupervised Domain Adaptation (UDA)

Fourier swapping

Semi-Supervised Learning (SSL)

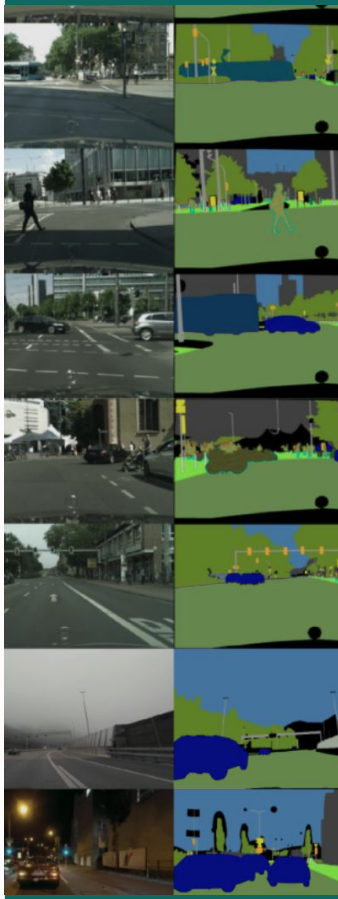
Challenge

Unlabeled data

Regularization model

Cluster assumption in the absence of Ground truth labels

Entropy Minimization



WHY ?

Robust Entropy Minimization

Problem

Standard entropy minimization (cross entropy loss) is often **ineffective** in regions where the model is already slightly certain but **wrong**.

$$\mathcal{L}_{ce}(\phi^w; D^{s\to t}) = -\sum_i \langle y_i^s, \log(\phi^w(x_i^{s\to t}))\rangle.$$

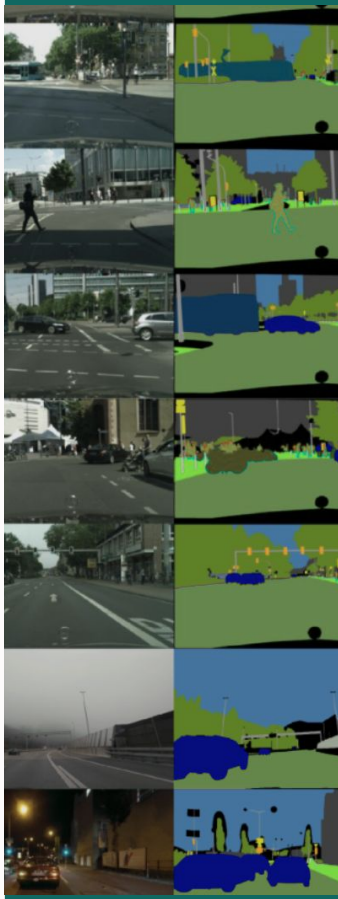
$\mathcal{L}_{ce}$: **Cross-Entropy Loss** (The primary training objective).

ϕ^w: **Segmentation Network** (The model and its trainable weights w).

$D^{s\to t}$: **Adapted Dataset** (Source images stylized to look like the target domain).

y_i^s: **Source Ground-Truth**

$x_i^{s\to t}$: **Transformed Input** (The synthetic image after the Fourier style swap).



WHY ?

Robust Entropy Minimization

Solution

Instead of an arbitrary threshold, we use the **Charbonnier penalty function:**

$$\rho(x) = (x^2 + 0.001^2)^\eta$$

η: The penalty curve.

The Entropy Minimization Loss:

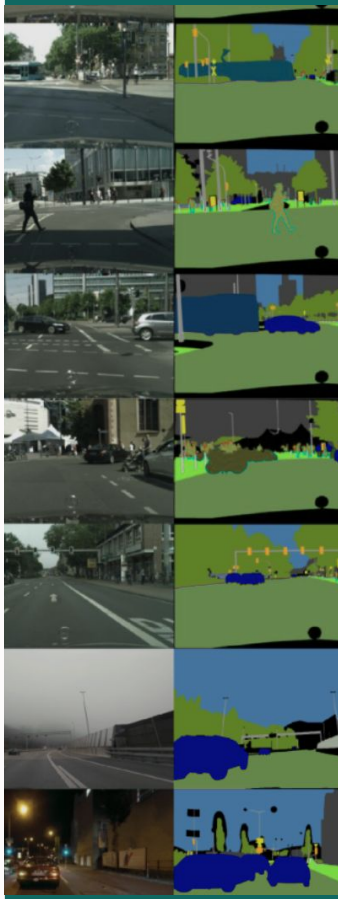
$$\mathcal{L}_{ent}(\phi^w; D^t) = \sum_i \rho(-\langle \phi^w(x_i^t), \log(\phi^w(x_i^t)) \rangle)$$

ϕ^w: **Segmentation Network** (The model predicting class probabilities).

x_i^t: **Target Input** (The real-world image without labels).

D^t: **Target Dataset** (The collection of real-world images without labels).

It measures certainty on the real-world target images.



WHY ?

Robust Entropy Minimization

The Total Loss

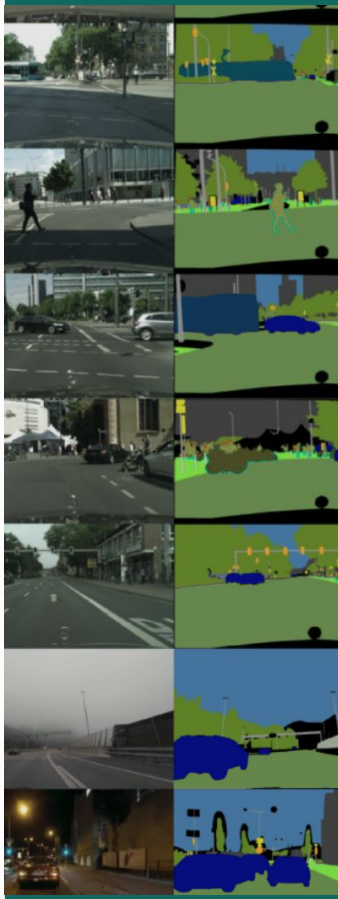
$$\mathcal{L}(\phi^w; D^{s\to t}, D^t) = \underbrace{\mathcal{L}_{ce}(\phi^w; D^{s\to t})}_{\text{Labeled data}} + \lambda_{ent}\underbrace{\mathcal{L}_{ent}(\phi^w; D^t)}_{\text{Unlabeled data}}$$

The model with **Charbonnier function**:

- Ignores very small amounts of uncertainty.
- Heavily punishes high uncertainty.

$\mathcal{L}_{ent}$ Reducing confusion on unlabeled data

$\mathcal{L}_{ce}$ Learning from labeled data



Multi-band Transfer (MBT)

In the FDA process, the parameter β controls the size of the spectral window being swapped between images.

Low β
The source image changes very little.

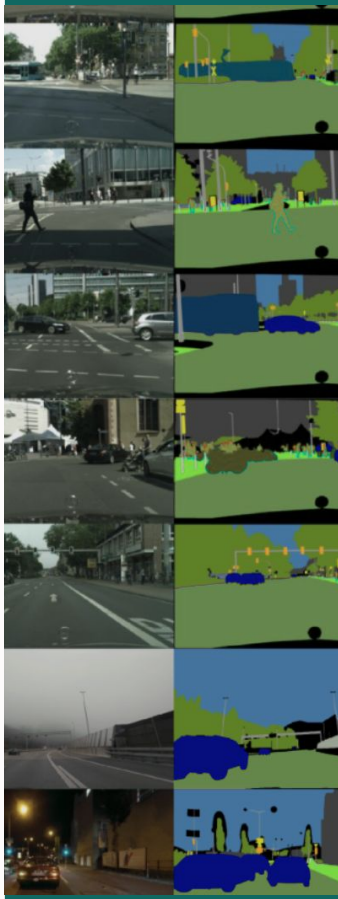
High β
Artifacts appear confusing the model.

Problem: no single β value is best for every object...

Values of β /IoU of a class(%)	Road	Car	Bus	mIoU(%)
β=0.01	88.8	81.7	**47.1**	44.61
β=0.05	90.7	**83.2**	36.9	44.6
β=0.09	**90.8**	82.8	44.1	45.01

Reminder

Intersection over Union (**IoU**) measures for a class, the ratio between the **intersected** area of **prediction** and **ground truth**, and the **combined** area of **prediction and ground truth.**



Multi-band Transfer (MBT)

Solution: Instead of training just one network, we instantiate **3 separate segmentation networks,** $(\phi^{\beta_1}, \phi^{\beta_2}, \phi^{\beta_3})$ and then we average their predictions.

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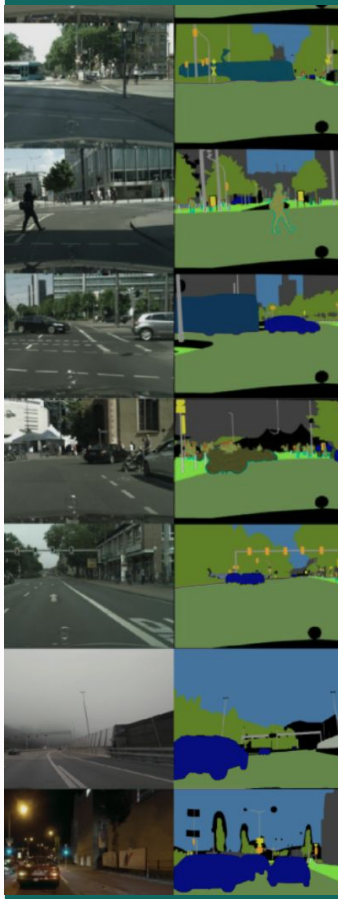
β=0.05	90.7	**83.2**	36.9	44.6
β=0.09	**90.8**	82.8	44.1	45.01
MBT (T=0)	**91.3**	**84.5**	**46.1**	**46.77**



Self-Supervised training (SST)

The final phase exploits the model's **own high-confidence** predictions on **real-world target images** to refine its **accuracy**.

Step A: Generating Pseudo-Labels for the unlabeled images.

- The ensemble makes a prediction for every pixel and then averages three different models (using MBT), so its guesses are very reliable.
- These guesses are called "**Pseudo-Labels**" $\hat{y}_i^t$. We treat them as if they were the "**Ground Truth**" provided by a human.

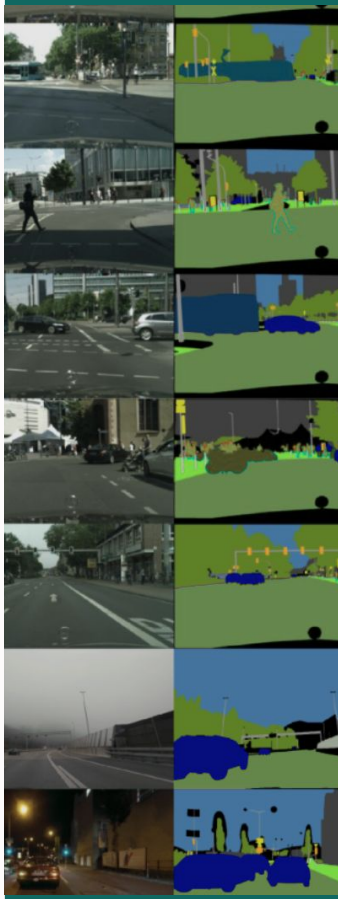
Step B: The "Denoising" Thresholds (The Quality Control)

To ensure **high quality**, the authors apply a strict **filter**. A pixel only gets a pseudo-label if:

- **Top 66% Confidence:** The model's confidence for that pixel is in the top portion of all pixels in that class.
- **Probability > 0.9:** The model is at least 90% sure about the class.

Step C: Iterative Training (factor T)

The model undergoes two additional rounds of training



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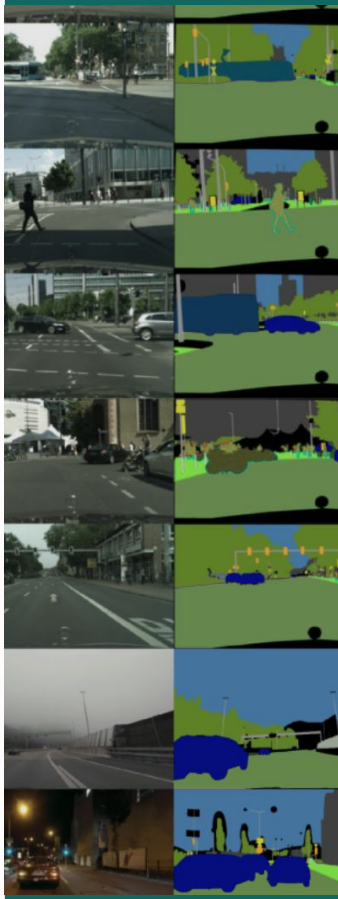
The final phase exploits the model's **own high-confidence** predictions on **real-world target images** to refine its **accuracy**.

Step A: Generating Pseudo-Labels for the unlabeled real-world images.

Step B: The "Denoising" Thresholds (The Quality Control)

Step C: Iterative Training (factor T)

Round of self-supervised training	mIoU(%)
MBT (T=0)	46.77
MBT (T=1)	48.14
MBT (T=2)	50.45



Experiments

Datasets

GTA5:

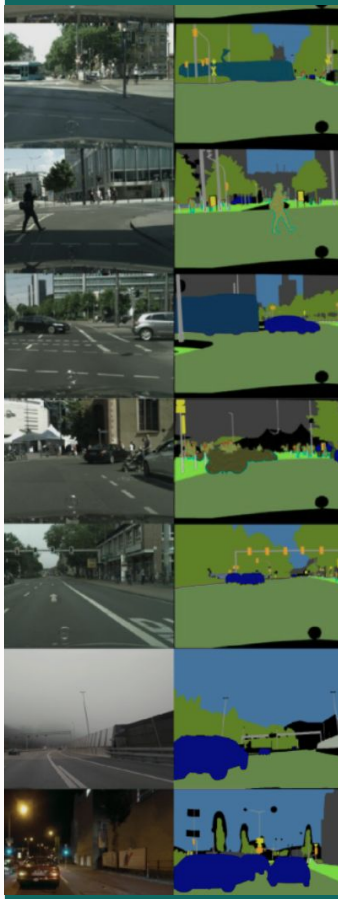
- 24,966 synthesized images captured in a video game, resized and random cropped to 1280×720.
- 19 classes used: these are in common with CityScapes for standard comparison to other state-of-the-art methods.

SYNTHIA: SYNTHIA-RAND-CITYSCAPES subset

- 9,400 annotated images with resolution randomly cropped to 1024×512 during training.
- 16 common classes are used for training.

CityScapes:

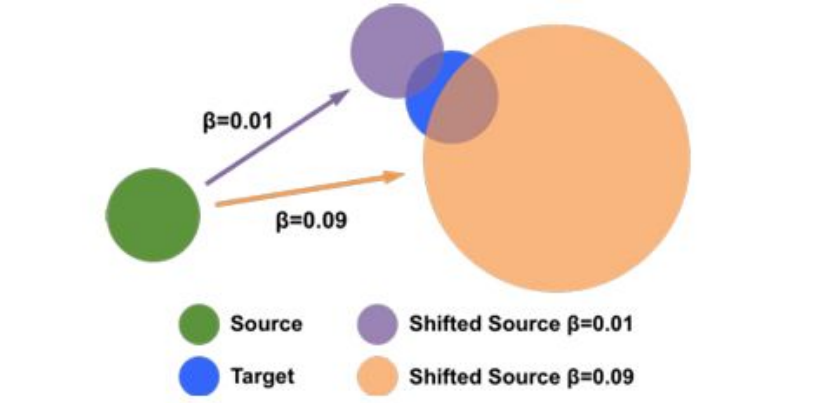
- 2,975 images from the training set as the target domain data for training.
- Test on 500 validation images.
- Images in CityScapes are resized to 1024×512, with no random cropping.



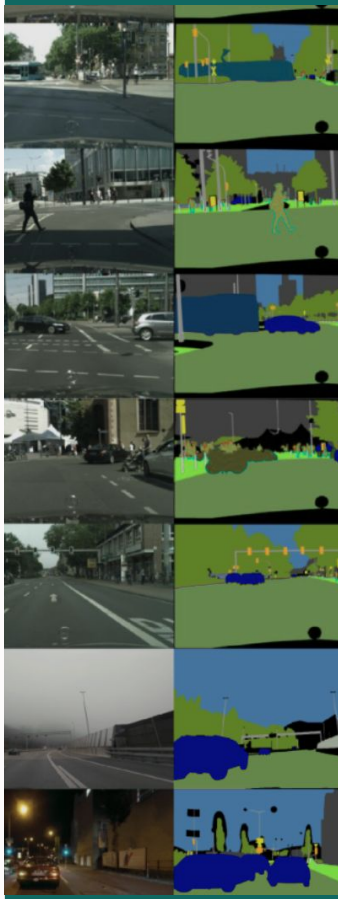
Ablation study on the GTA5->CityScapes task

The segmentation networks trained with different β's in the FDA operation maintain similar performance.

FDA is robust with respect to the choice of β. Without entropy loss, with β=0.09, it surpasses the baseline Cycada by 4.54%

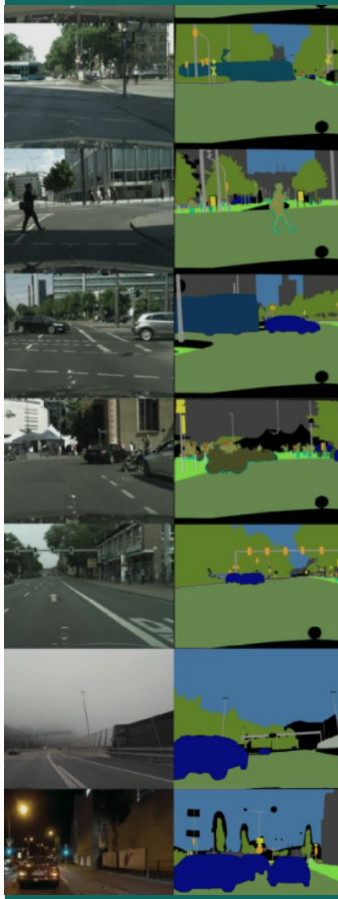


Using SST with MTB: best performance in the training from scratch and worst performance in the first round of SST with β=0.09.
After the second round of SST: best performance with β=0.01



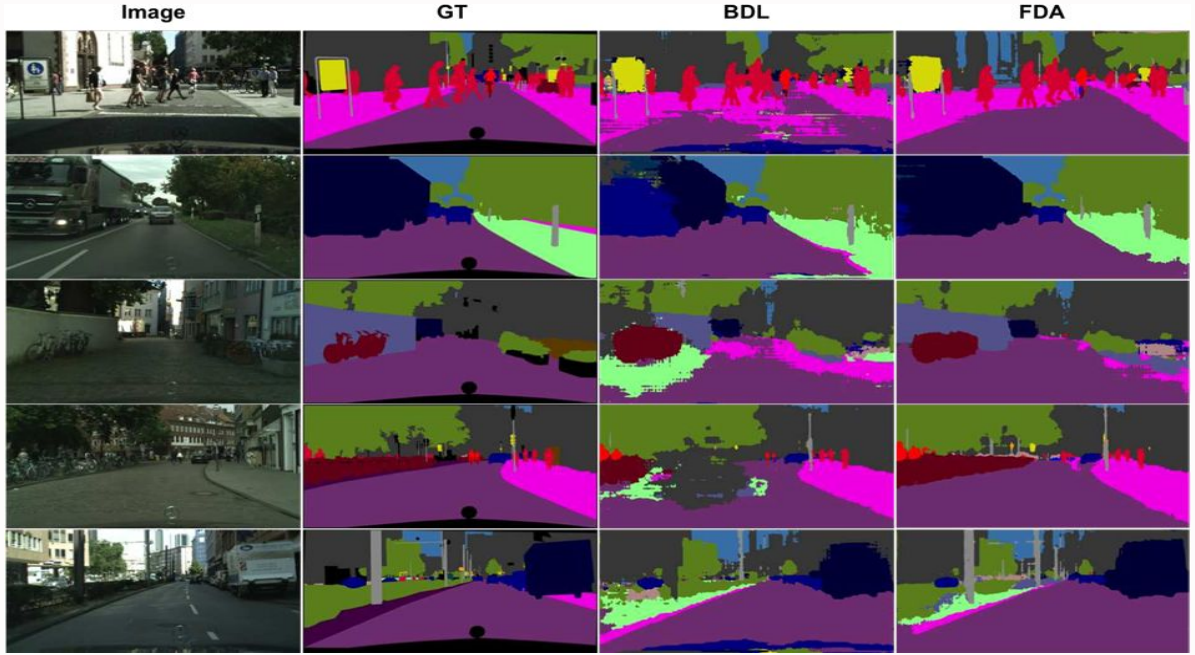
Different mIoU across different methods

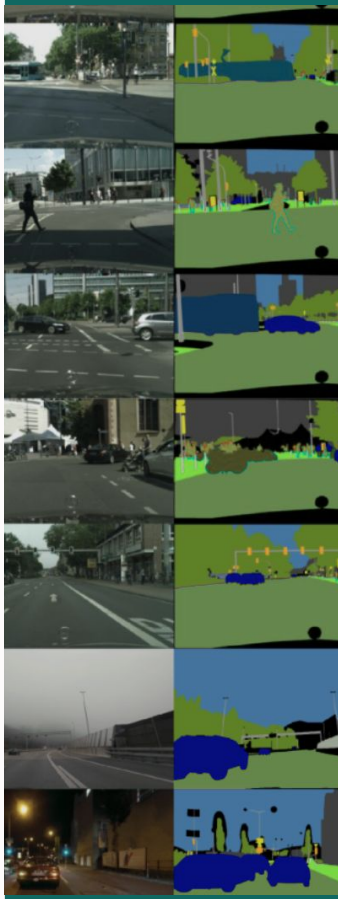
Backbone	Datasets	Method	MIoU
ResNet101	GTA5->CityScapes	BDL	48.5
ResNet101	GTA5->CityScapes	**FDA-MBT**	50.45 **(+4%)**
VGG16	GTA5->CityScapes	BDL	41.3
VGG16	GTA5->CityScapes	**FDA-MBT**	42.2 **(+2.13%)**
ResNet101	SYNTHIA->CityScapes	BDL	51.4
ResNet101	SYNTHIA->CityScapes	**FDA-MBT**	52.5 **(+2.1%)**
VGG16	SYNTHIA->CityScapes	BDL	39.0
VGG16	SYNTHIA->CityScapes	**FDA-MBT**	40.5 **(+3.9%)**



Visual comparison with another perfomer

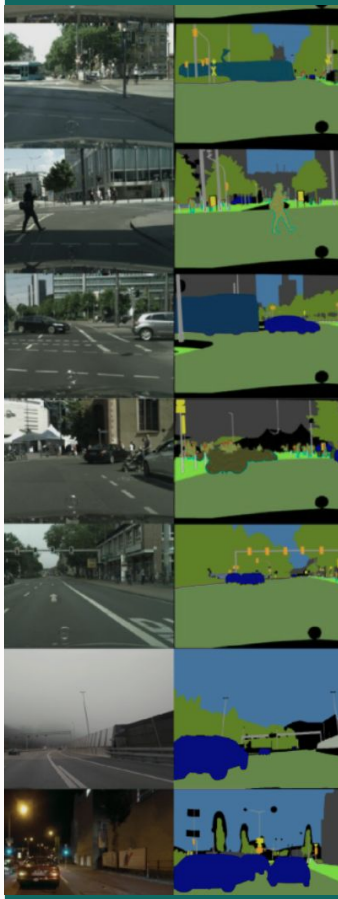
Predictions appear **much less noisy and smoother**. FDA maintains the **fine structures** and performs well on **rare classes** (truck and bicycles).
This is due to both **generalization ability** of the **single scale FDA** and the **regularized SST by MBT**.





Conclusions

- FDA **surpasses** the current state of the art.
- It may **not** be necessary to **learn what we already know** (low-level statistics of the image can vary without affecting the semantics of the underlying scene).
- This preprocessing could be an **alternative** to sophisticated architectures or laborious data augmentation.
- **Robustness to nuisance variability** affecting the image domain remains a **difficult problem** in machine learning.



THANK YOU FOR YOUR ATTENTION!